Package ‘flexTab1’

May 29, 2026

Title Creation of a Table 1 and descriptive summary tables

Version 0.0.0.9000

Description This function creates a highly flexible summary measure table for descriptive statistics (``Table 1") with options to customize both the statistical options and the output layout.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

URL https://github.com/KGutmair/flexTab1

BugReports https://github.com/KGutmair/flexTab1/issues

Imports checkmate (>= 2.3.1),
flextable (>= 0.9.6),
magrittr (>= 2.0.3),
smd (>= 0.7.0),
stringr (>= 1.5.1),
rlang (>= 1.1.4)

Suggests knitr,
rmarkdown,
survival,
officer,
dplyr

VignetteBuilder knitr

R topics documented:

`Table1_flex` *Generate Table 1 with Customizable Summary Measures*

Description

This function creates a highly flexible Table 1 for descriptive statistics. It allows users to customize the output, choose summary measures, display missing values, p-values (testing for difference between two groups),and standardized mean differences (SMD).

Usage

```
Table1_flex(
  data,
  variables,
  group_var = FALSE,
  treatment_arm = FALSE,
  new_line = FALSE,
  measures_cat = c("absolute", "relative"),
  measures_num = c("median", "min", "max"),
  display_pvalue = FALSE,
  display_smd = FALSE,
  display_missings = TRUE,
  flextable_output = TRUE,
  sort_rows = NULL,
  add_measure_ident = TRUE,
  treatment_order = NULL,
  group_order = NULL
)
```

Arguments

`data`	A `data.frame` containing the data for which the Table 1 should be generated
`variables`	A character vector specifying the variable names for which the Table 1 summary should be created.
`group_var`	A character string specifying the grouping variable for stratification. If set to `FALSE` (default), the Table 1 is not stratified by groups.
`treatment_arm`	A character string indicating the main grouping variable. Subgroups defined by group_var are nested within each level of this variable. For example, in a dataset with three treatment arms ("treatment": A, B, C) and two subgroups within each treatment category ("sub_group": I, II), set `group_var` to "sub_group" and `treatment_arm` to "treatment". If p-values or SMD options are selected, the subgroups within the treatment groups are compared. Default is `FALSE` if no treatment arms are used.
`new_line`	logical (TRUE/FALSE), default is `FALSE`. If `TRUE`, an empty row is added after every variable in the table, which can improve clarity, especially when returning the Table 1 as a flextable object.
`measures_cat`	A character vector indicating how categorical variables should be summarized. Options are `"absolute"`, `"relative"`, or both. Default is `c("absolute", "relative")` for displaying both absolute and relative frequencies.

measures_num
: A character vector defining the summary statistics for numeric variables. Options include "median", "min", "max", "quantile1" (25th percentile), "quantile3" (75th percentile), "mean", and "sd" (standard deviation). The number of measures is limited to three (e.g., mean(sd), median(min-max), or median(quantile1-quantile3)).

display_pvalue
: logical (TRUE/FALSE), default is FALSE. If TRUE, p-values are displayed.

display_smd
: logical (TRUE/FALSE), default is FALSE. If TRUE, standardized mean differences (SMD) is displayed.

display_missings
: A logical (TRUE/FALSE), default is FALSE. If TRUE, the frequency of missing values for each variable is shown in the output.

flextable_output
: A logical (TRUE/FALSE), default is TRUE. If TRUE, the Table 1 is returned as a flextable object. If FALSE, it is returned as a data.frame.

sort_rows
: A character vector specifying the row order in which the variables should appear in the output table. The default is NULL, which sorts the variables alphabetically.

add_measure_ident
: A logical (TRUE/FALSE), default is TRUE. If TRUE, the summary measures identifiers, e.g. median (min-max), mean (sd), n (%) is added to the Table 1.

treatment_order
: character vector containing the categories of the treatment_arm parameter in the desired column order in the output table.

group_order
: character vector containing the categories of the group_var parameter in the desired column order in the output table.

Details

Testing for differences in variables between groups. For numerical variables, a Wilcoxon rank-sum test is used to assess differences between the median of two groups at a significance level of 5\ For categorical variables, a Chi-square test is used to assess differences between groups at a significance level of 5\ absolute count is equal or below 5, then the Fisher's exact test is used instead. Till now, only group comparisons between two groups is implemented.

Value

A data.frame or flextable object, depending on the selected output option. The returned Table 1 contains the summary measures specified in the input parameters, including any chosen statistics, p-values, standardized mean differences (SMD), and missing value frequencies.

References

Austin, P. C. (2011). An Introduction to Propensity Score Methods for Reducing the Effects of Confounding in Observational Studies. Multivariate Behavioral Research, 46(3), 399–424. https://doi.org/10.1080/00273171

Examples

```
if (requireNamespace("survival", quietly = TRUE)) {
  # Load pbc data from the survival package
  pbc <- survival::pbc

  baseline_var <- c("age", "chol", "sex", "stage", "platelet")
```

```
  pbc[c("stage", "trt", "edema", "hepato")] <-
    lapply(pbc[c("stage", "trt", "edema", "hepato")], as.factor)

  pbc <- pbc[!is.na(pbc$trt), ]
pbc$trt <- ifelse(pbc$trt == "1",
"D-penicillamine",
"Placebo")

# Table 1 for one group ------------------------------------------

tab1_ex <- Table1_flex(
data = pbc,
variables = baseline_var
)
flextable::autofit(tab1_ex)

# choosing other summary measures
Table1_flex(
data = pbc,
variables = baseline_var,
measures_num = c("mean", "sd"),
measures_cat = "relative"
)

# Returning a data.frame for further processing
# and other summary measures
Table1_flex(
data = pbc,
variables = baseline_var,
measures_num = c("mean", "sd"),
measures_cat = c("absolute","relative"),
flextable_output = FALSE
)

# Table 1 for two groups -----------------------------------------

tab1_ex <- Table1_flex(
data = pbc,
variables = baseline_var,
group_var = "trt",
add_measure_ident = FALSE,
new_line = TRUE,
sort_rows = c("age", "sex", "stage", "chol", "platelet"),
group_order = c("Placebo", "D-penicillamine")
)

flextable::autofit(tab1_ex)

# with additionally p-values and standarized mean differences
tab1_ex <- Table1_flex(
data = pbc,
variables = baseline_var,
group_var = "trt",
add_measure_ident = FALSE,
new_line = TRUE,
sort_rows = c("age", "sex", "stage", "chol", "platelet"),
```

```
group_order = c("Placebo", "D-penicillamine"),
display_pvalue = TRUE,
display_smd = TRUE
)

flextable::autofit(tab1_ex)


# Table 1 for a nested group structure ----------------------------

baseline_var <- c("age", "chol", "platelet", "stage")
tab1_ex <- Table1_flex(
data = pbc,
variables = baseline_var,
group_var = "sex",
treatment_arm = "trt",
add_measure_ident = TRUE,
sort_rows = c("age", "stage", "chol", "platelet"),
flextable_output = TRUE,
display_pvalue = TRUE
)

flextable::autofit(tab1_ex)
}
```

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